FACULTY OF ENGINEERING · UNIVERSITY OF PERADENIYA · CO2060 RESEARCH PROJECT · TIMESWARE

Clinical Staff Feedback Form

Automated PAR Index Calculation System

All responses are strictly confidential

PROJECT
CO2060 · 2nd year

CONTACT
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RESPONDENT
Clinical / Academic Staff

Purpose: This form collects your expert clinical opinion on the usability, accuracy, and practical value of our web-based system for automated PAR Index calculation. Your input will directly inform our development priorities for 2nd year and shape the system's readiness for real-world clinical deployment.

RESPONDENT SIGNATURE

DATE
2026/02/22

1 Respondent Information

CLINICAL ROLE
Orthodontist

YEARS OF CLINICAL EXPERIENCE
2026

INSTITUTION / HOSPITAL
Dental hospital

EMAIL ADDRESS (OPTIONAL — FOLLOW-UP ONLY)
Peradeniya

Have you used any similar digital orthodontic assessment tools before?

- [ ] Yes
- [x] No
- [ ] Unsure

IF YES, PLEASE NAME THE SYSTEM(S)

2 General Usability

Rate your level of agreement · 1 = Strongly Disagree · 5 = Strongly Agree

Statement	1 Str. Disagree	2 Disagree	3 Neutral	4 Agree	5 Str. Agree
The overall interface is intuitive and easy to navigate.					✓
I was able to understand the system's output without additional guidance.					✓
The workflow matches how I approach PAR scoring in clinical practice.			✓		
The system minimises unnecessary complexity for clinical staff.					✓
I would feel comfortable using this system with minimal prior training.					✓

3 Feature-Specific Evaluation

Rate the quality / usefulness of each feature · 1 = Very Poor · 5 = Excellent

Feature	1 Very Poor	2 Poor	3 Fair	4 Good	5 Excellent
File Upload — ease of uploading 3D models (.STL / .OBJ) and file validation feedback					✓
Automated Landmark Detection — accuracy and reliability of AI-identified anatomical points			✓		
3D Visualisation Viewer — rendering quality, zoom / rotate / pan responsiveness					✓
Manual Override Tool — ability to adjust and correct AI-placed landmark positions				✓	
PDF Report Generation — clarity, completeness, and clinical usefulness of the output report					✓

4 Clinical Assessment & Confidence

4.1

How easy was it to upload a 3D dental model and initiate PAR calculation?

Very Difficult | Difficult | Neutral | Easy | Very Easy

4.2

How would you rate the 3D model viewer's performance and visual clarity?

Very Poor | Poor | Acceptable | Good | Excellent

4.3

How useful is the manual landmark override feature for clinical practice?

Not useful at all | Slightly useful | Moderately useful | Very useful | Essential

4.4

How suitable are the generated PDF reports for inclusion in clinical records?

Not suitable | Needs major revision | Needs minor revision | Mostly suitable | Fully suitable

4.5

How confident are you in the clinical accuracy of the AI-computed PAR scores? (1 = Not at all confident · 10 = Extremely confident)

1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10

Not at all confident *Extremely confident*

4.6

What concerns, if any, do you have about integrating AI-based PAR scoring into clinical workflows? (Tick all that apply)

- [x] Accuracy and reliability of AI landmark detection
- [ ] Liability and accountability for AI-generated results
- [ ] Data privacy and patient confidentiality
- [ ] Over-reliance on automation reducing clinical skill
- [ ] System errors causing incorrect treatment decisions
- [ ] Resistance or lack of trust from the clinical team

- [ ] No major concerns

5 Efficiency & Time Savings

5.1

Compared to manual PAR scoring, how much time do you estimate this system saves per case?

No time saved | Less than 5 min | 5–10 minutes | 10–15 minutes | More than 15 min

5.2

Which stage of the digital workflow felt the most time-consuming?

File Upload | AI Landmark Detection | Manual Correction / Override | Report Export | None — all stages were fast

6 Clinical Validation & Error Analysis

6.1

In approximately what percentage of cases did you feel the need to use the Manual Override tool?

- 0% — AI was always accurate
- Less than 25%
- Around 50%
- More than 75%
- Used override on every case

6.2

Are there specific dental landmarks where the AI consistently underperforms? (Tick all that apply)

- [ ] Upper anterior segment / overjet
- [ ] Lower anterior segment
- [ ] Right buccal occlusion (molar relationship)
- [ ] Left buccal occlusion (molar relationship)
- [ ] Overbite / vertical overlap
- [ ] Centreline / midline deviation
- [ ] No consistent underperformance noticed

6.3

How closely did the AI-generated PAR score match your own manual calculation for the same model? (1 = Completely different · 10 = Perfect match)

1 2 3 4 5 6 7 8 9 10

Completely different *Perfect match*

7 Technical & System Performance

7.1

How would you rate the system's responsiveness when manipulating the 3D model (zoom, rotate, pan)?

- Very laggy / unusable
- Noticeable lag
- Acceptable
- Smooth
- Very smooth / responsive

7.2

If a file upload failed or a calculation crashed, was the error feedback from the system helpful enough to resolve the issue?

- Did not encounter any errors
- Error message was confusing
- Partially helpful
- Mostly helpful
- Fully helpful — resolved immediately

8 Integration & Compatibility

8.1

Did you experience any lag or rendering issues on the device you were using?

No issues at all | Minor lag only | Moderate lag | Significant lag | System was unusable on my device

Device used:

Laptop | Desktop PC | Tablet | Other

IF 'OTHER', SPECIFY DEVICE

8.2

How important is it for this system to sync directly with your existing Patient Management System (PMS)?

Not important | Slightly important | Moderately important | Very important | Essential — would not adopt without this

9 PDF Report Utility

9.1

Does the PDF report provide enough visual evidence (e.g. landmark screenshots) to justify the PAR score to a third party or auditor?

Definitely not | Probably not | Unsure | Probably yes | Definitely yes

9.2

How effective would this report be as a visual aid when explaining treatment needs to a patient?

Not effective at all | Slightly effective | Moderately effective | Very effective | Extremely effective

10 Overall Assessment

10.1

Overall, how would you rate this system for clinical use?

Very Poor | Poor | Fair | Good | Excellent

10.2

Would you recommend this system to colleagues or other orthodontic institutions?

Definitely not | Probably not | Unsure | Probably yes | Definitely yes

10.3

Which features do you consider most valuable for clinical practice? (Tick all that apply)

- [x] Automated landmark detection
- [ ] 3D model visualisation
- [ ] Manual override and correction
- [ ] Longitudinal patient tracking
- [ ] Downloadable PDF reports
- [ ] Model versioning and reproducibility
- [ ] Multi-institution / cloud access

11 Open-Ended Feedback

11.1

What specific improvements would most increase your confidence in using this system clinically?

need automated PAR score generation.

11.2

Are there any clinical or institutional barriers that might prevent adoption of this system at your facility?
(e.g. IT infrastructure, ethical clearance, data governance, training requirements)

NO, for this no need or any ~~[illegible]~~ monetary allocation.

11.3

If you could automate one additional part of your clinical workflow (e.g. cephalometric analysis, treatment outcome prediction), what would be your top priority?

cephalometric analysis.

11.4

Any additional comments, suggestions, or features you would like to see in future versions?

This is a really a good work. but without automation accuracy we cannot impliment this.